

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iv)	Obtaining $Q_0 = 19.1 \times 2.2 \text{ mF} = 42 \pm 2 \text{ m[C]}$ (1) Using 63 % of fully charged OR substituting values into $Q = Q_0 \left(1 - e^{-\frac{t}{RC}}\right)$ OR taking logs correctly (1) T or $RC = 27 \pm 2 \text{ [s]}$ (no marks for 26.4 s) OR Q value correct after substituting (1)		3		3	3	3
		(v)	Good tangent drawn at 45 s ecf (1) Gradient = 0.27 m[A] ecf (1)		2		2		2
	(c)		Any 5 × (1) from: - Shape of graph is good - Line of best fit goes through error bar(s) - $RC = 26.4 \text{ s}$ - RC in good agreement with 27 s (ecf available) - $I_0 = \frac{19.1}{12\,000} = 1.59 \text{ m[A]}$ - Gradient of graph decreases / gradient is the current - So current equation is in good agreement – only award mark if current decreasing implied - Substitution into either equation for confirmation			5	5	2	5
			Question 1 total	4	15	8	27	8	18